

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIC501 - COMPUTER NETWORK AND CLOUD COMPUTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 **What is the primary use of computer networks?**
 - a) Performing arithmetic operations
 - b) Data storage
 - c) Resource sharing and communication
 - d) Video editing
- 2 **Which of the following is a reference model in networking?**
 - a) TCP b) OSI c) HTTP d) IP
- 3 **Network standardization is important because:**
 - a) It slows down innovation
 - b) It limits hardware use
 - c) It ensures interoperability and compatibility
 - d) It prevents network attacks
- 4 **Software-defined networking (SDN) separates which two planes?**
 - a) Physical and virtual b) Control and data
 - c) Wired and wireless d) Cloud and network
- 5 **Which protocol is used for email transmission?**
 - a) HTTP b) FTP c) SMTP d) DNS
- 6 **A benefit of cloud computing is:**
 - a) High upfront investment b) No scalability
 - c) On-demand resource access d) Limited data sharing
- 7 **SLAs in cloud computing refer to:**
 - a) Security License Agreement b) Service Level Agreements
 - c) Software License Activation d) System Level Adjustment
- 8 **Amazon EC2 is an example of:**
 - a) SaaS b) PaaS c) IaaS d) CaaS

- 9 **Software-as-a-Service (SaaS) provides:**
 a) Full virtual machines
 b) Complete software applications over the internet
 c) Network cables
 d) Storage devices
- 10 **Which cloud service model allows users to deploy and manage applications without dealing with underlying infrastructure?**
 a) IaaS b) PaaS c) SaaS d) CaaS
- What is the main purpose of the transport layer?**
 a) IP addressing b) Secure login
 c) Reliable data transfer d) Routing
- The purpose of routing algorithms is to:**
 a) Control audio volume
 b) Decide the best path for data transmission
 c) Block IP addresses
 d) Encrypt data
- Q2. Solve the following.
- A) Explain the OSI and TCP/IP reference models. 6
- B) Discuss how network technologies have evolved from local to global networks. 6
- Q3. Solve the following.
- A) Discuss the concept and architecture of Software-Defined Networking (SDN). 6
- B) What are the design issues in the data link layer? 6
- Q4. Solve Any Two of the following.
- A) What are the key responsibilities of the transport layer in the OSI and TCP/IP models? 6
- B) Differentiate TCP & UDP in detail. 6
- C) How does the HTTP protocol work in the context of web communication? 6
- Q5. Solve Any Two of the following.
- A) How does cloud computing differ from traditional computing paradigms like client-server? 6
- B) Compare the cloud deployment models: public, private, hybrid, and community clouds. 6
- C) What are the essential components of a modern data center? How do they support cloud operations? 6
- Q6. Solve Any Two of the following.
- A) What is Communication-as-a-Service (CaaS), and how does it facilitate real-time collaboration in cloud-based environments? 6
- B) Explain Software-as-a-Service (SaaS) with examples of widely-used applications. 6
- C) What is Amazon EC2 and how does it implement IaaS principles? 6

025339761

025339761

025339761

025339761

025339761

025339761

025339761

025339761

025339761

*** End ***